



OPERATIONAL RESEARCH & OPTIMISATION

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Outline

- Introduction to Multi-Objective Optimization Problems (MOOPs)
- Concept of Pareto Optimal Solution
- Goal Programming approach to MOOPs

Multi-Objective Optimization Problems (MOOP)

- Involve more than one objective function that are to be minimized or maximized
- Answer is set of solutions that define the best tradeoff between competing objectives

General Form of MOOP

■ Mathematically

$$\min/\max f_m(\mathbf{x}), \quad m=1, 2, \dots, M$$

$$\text{subject to } g_j(\mathbf{x}) \geq 0, \quad j=1, 2, \dots, J$$

$$h_k(\mathbf{x}) = 0, \quad k=1, 2, \dots, K$$

$$\underset{\text{lower bound}}{x_i^{(L)}} \leq x_i \leq \underset{\text{upper bound}}{x_i^{(U)}}, \quad i=1, 2, \dots, n$$

Dominance

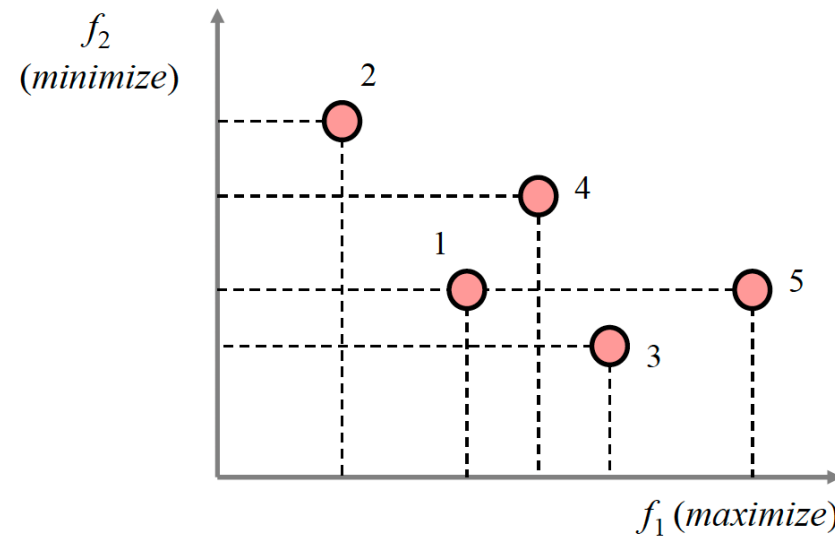
- In the single-objective optimization problem, the superiority of a solution over other solutions is easily determined by comparing their objective function values
- In multi-objective optimization problem, the goodness of a solution is determined by the **dominance**

Definition of Dominance

■ Dominance Test

- x_1 dominates x_2 , if
 - Solution x_1 is no worse than x_2 in all objectives
 - Solution x_1 is strictly better than x_2 in at least one objective
- x_1 dominates $x_2 \iff x_2$ is dominated by x_1

Example Dominance Test

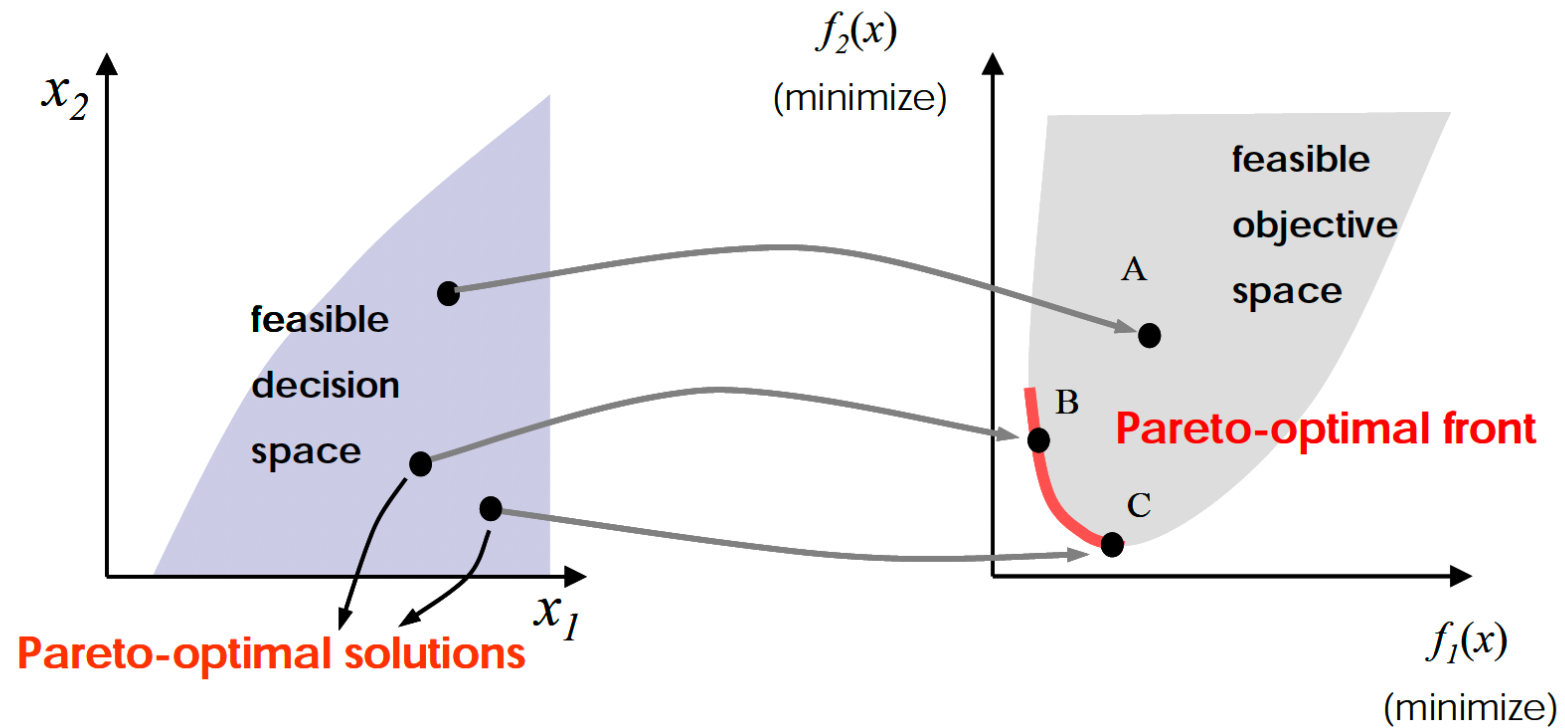


- 1 Vs 2: 1 dominates 2
- 1 Vs 5: 5 dominates 1
- 1 Vs 4: Neither solution dominates

Pareto Optimal Solution

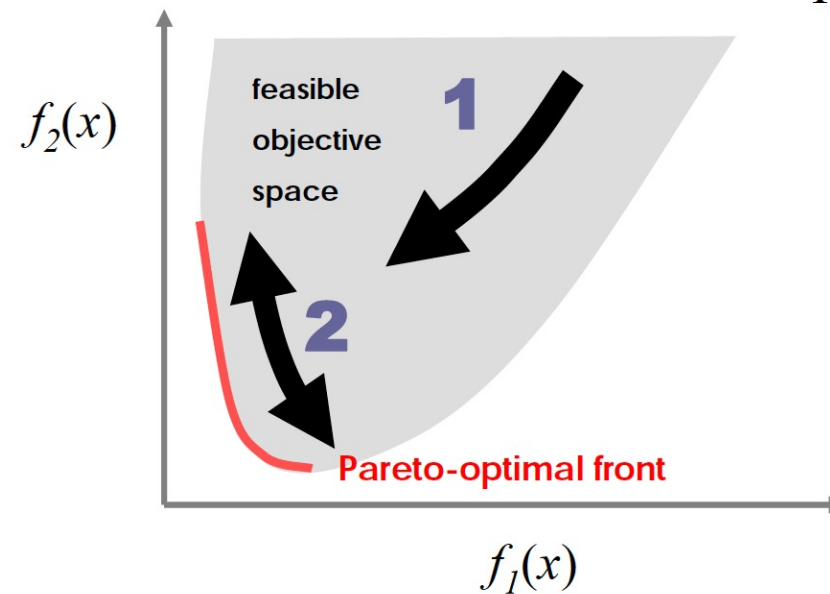
- **Non-dominated solution set**
 - Given a set of solutions, the non-dominated solution set is a set of all the solutions that are not dominated by any member of the solution set
- The non-dominated set of the entire feasible decision space is called the **Pareto-optimal set**
- The boundary defined by the set of all point mapped from the Pareto optimal set is called the **Pareto-optimal front**

Graphical Depiction of Pareto Optimal Solution



Goals in MOO

- Find set of solutions as close as possible to Pareto-optimal front
- To find a set of solutions as diverse as possible



Classic MOO methods

- Weighted Sum Method
- (Weighted Metric Method)(out of scope)
- (Multi-Objective Evolutionary Algorithms)(out of scope)
- Goal Programming approach

Weighted Sum Method

- Scalarize a set of objectives into a single objective by adding each objective pre-multiplied by a user-supplied weight

$$\begin{aligned} \text{minimize} \quad & F(\mathbf{x}) = \sum_{m=1}^M w_m f_m(\mathbf{x}), \\ \text{subject to} \quad & g_j(\mathbf{x}) \geq 0, \quad j = 1, 2, \dots, J \\ & h_k(\mathbf{x}) = 0, \quad k = 1, 2, \dots, K \\ & x_i^{(L)} \leq x_i \leq x_i^{(U)}, \quad i = 1, 2, \dots, n \end{aligned}$$

- Weight of an objective is chosen in proportion to the relative importance of the objective

Weighted Sum Method

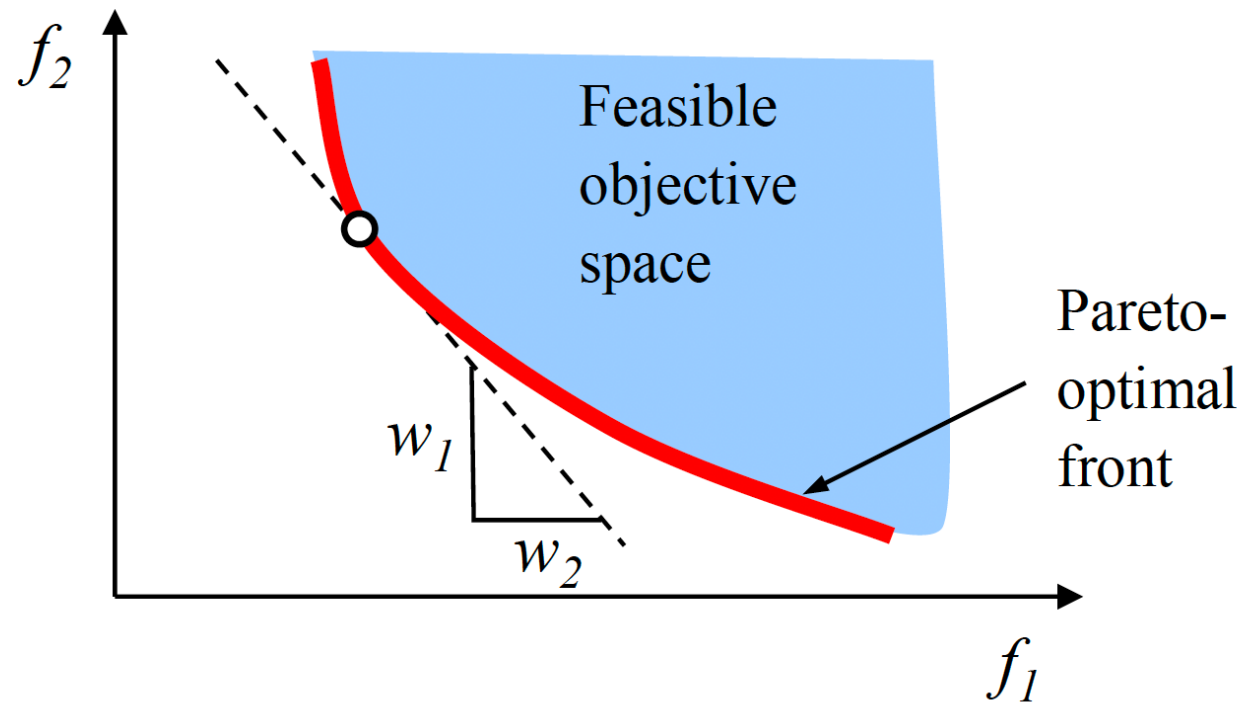
- Advantage

- Simple

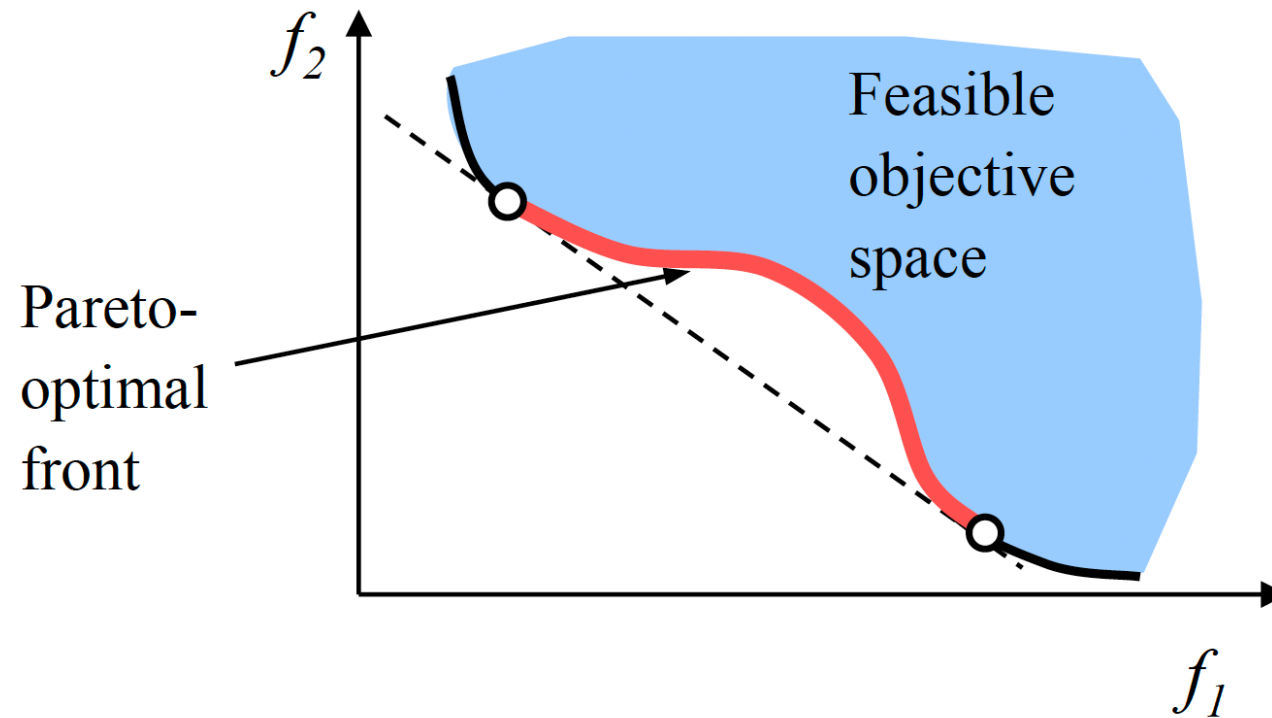
- Disadvantage

- It is difficult to set the weight vectors to obtain a Pareto-optimal solution in a desired region in the objective space
 - It cannot find certain Pareto-optimal solutions in the case of a nonconvex objective space

Weighted Sum Method (Convex Case)



Weighted Sum Method (Non-Convex Case)



How to construct Pareto optimal frontier?

- See Case study in LW7

Goal Programming

- Goal programming (GP) is a well-established technique for solving **MOOPs**.
- It is particularly useful when you have multiple, often conflicting objectives, and you want to find a solution that best satisfies a set of **goals or targets** for each objective.

How GP works

- 1. Define Goals:** For each objective, you specify a target or goal value that you want to achieve.
- 2. Deviations:** Goal programming allows for deviations (both positive and negative) from these goals. The aim is to minimize the weighted sum of these deviations.
- 3. Prioritization:** You can assign priorities or weights to the goals to reflect their relative importance.
- 4. Formulate the Problem:** The problem is transformed into a single-objective optimization problem where the objective is to minimize the total deviation from the goals.

An example-Production mix

Production mix LP formulation:

Let X_1 = number of chandeliers produced

X_2 = number of ceiling fans produced

Maximize profit $Z = 7X_1 + 6X_2$

subject to $2X_1 + 3X_2 \leq 12$ (wiring hours)

$6X_1 + 5X_2 \leq 30$ (assembly hours)

$X_1, X_2 \geq 0$

An example-Production mix

- Moving to a new location and maximizing profit is not a realistic objective
- A profit level of \$30 would be satisfactory during this period
- The goal programming problem is to find the production mix that achieves this goal as closely as possible given the production time constraints
- Define two deviational variables
 - d_1^- = underachievement of the profit target
 - d_1^+ = overachievement of the profit target

An example-Production mix

Single-goal programming formulation

Minimize under or overachievement of profit target $Z = d_1^- + d_1^+$

subject to

$$7X_1 + 6X_2 + d_1^- - d_1^+ = 30 \quad (\text{profit goal constraint})$$

$$2X_1 + 3X_2 \leq 12 \quad (\text{wiring hours})$$

$$6X_1 + 5X_2 \leq 30 \quad (\text{assembly hours})$$

$$X_1, X_2, d_1^-, d_1^+ \geq 0$$

An example-Production mix

- Note:
 - Analyze each goal to see if underachievement or overachievement of that goal is acceptable
 - If overachievement is acceptable, eliminate the appropriate d^+ variable from the objective function
 - If underachievement is okay, the d^- variable should be dropped
 - If a goal must be attained exactly, both d^- and d^+ must appear in the objective function

Extension to Equally Important Multiple Goals

Achieve several goals that are equal in priority

- Goal 1:* to produce a profit of \$30 if possible during the production period
- Goal 2:* to fully utilize the available wiring department hours
- Goal 3:* to avoid overtime in the assembly department
- Goal 4:* to meet a contract requirement to produce at least seven ceiling fans

Extension to Equally Important Multiple Goals

The deviational variables can be defined as

d_1^- = underachievement of the profit target

d_1^+ = overachievement of the profit target

d_2^- = idle time in the wiring department (underutilization)

d_2^+ = overtime in the wiring department (overutilization)

d_3^- = idle time in the assembly department (underutilization)

d_3^+ = overtime in the assembly department (overutilization)

d_4^- = underachievement of the ceiling fan goal

d_4^+ = overachievement of the ceiling fan goal

Extension to Equally Important Multiple Goals

Management is unconcerned about d_1^+ , d_2^+ , d_3^- , and d_4^+ so these may be omitted from the objective function

New objective function and constraints

Minimize total deviation $Z = d_1^- + d_2^- + d_3^+ + d_4^-$

subject to

$$\begin{aligned}
 \$7X_1 + \$6X_2 + d_1^- - d_1^+ &= \$30 \text{ (profit constraint)} \\
 2X_1 + 3X_2 + d_2^- - d_2^+ &= 12 \text{ (wiring hours constraint)} \\
 6X_1 + 5X_2 + d_3^- - d_3^+ &= 30 \text{ (assembly hours constraint)} \\
 X_2 + d_4^- - d_4^+ &= 7 \text{ (ceiling fan constraint)} \\
 \text{All } X_i, d_i \text{ variables} &\geq 0
 \end{aligned}$$

Goal Programming with Weighted Goals

- Weights can be used in the objective to allow the decision maker to indicate the relative importance of various goals
- Minimize total deviation $Z = w_1d_1^- + w_2d_2^- + w_3d_3^+ + w_4d_4^-$
- This approach is called **Non-preemptive** Goal Programming.
- Preemptive GP is supplied as additional reading.

Comments

Advantages of Goal Programming:

- It provides a flexible framework for handling multiple objectives.
- It allows for trade-offs between objectives.
- It can incorporate decision-maker preferences through weights or priorities.

Limitations:

- The choice of goals and weights can be subjective.
- It may not always find Pareto-optimal solutions.

- Q&A